Factory Machine Failure Detection

Telemetry Downtime Analysis for Daikibo Manufacturing

Deloitte Australia — Data Analytics Job Simulation (Forage) | Task: Telemetry Downtime Dashboard | Tool: Tableau Public | Author: Jay

1. Project Overview

Daikibo Manufacturing operates factories across three countries and wanted to understand where machine downtime was concentrated across its global operations. As part of this simulation, I acted as a data analyst tasked with taking a raw telemetry export from four factories and turning it into a clear, decision-ready view of where operational risk was highest — by factory and by machine type — so that the maintenance and operations teams could prioritise where to intervene first.

Devices across all sites report a status of **healthy** or **unhealthy** roughly every 10 minutes. An **unhealthy** reading is treated as a proxy for 10 minutes of potential downtime since the previous reading, which converts a simple status log into an estimate of lost operating time.

2. Dataset

Attribute	Value
Source file	daikibo-telemetry-data.json
Total telemetry readings	160,704
Reporting interval	~10 minutes per device
Time span covered	30 Apr 2021 – 09 May 2021 (~9 days)
Factories	4 (daikibo-berlin, daikibo-factory-meiyo, daikibo-factory-seiko, daikibo-shenzhen)
Countries	Germany, Japan, China
Devices monitored	36 (9 per factory)
Device types	AirWrench, CNC, ConveyorBelt, Furnace, HeavyDutyDrill, LaserCutter, LaserWelder, MetalPress, SpotWelder
Fields per reading	deviceID, deviceType, timestamp, location (country/city/area/factory/section), status, temperature

Every factory runs an identical fleet: the same 9 device types, one of each, per site — so downtime comparisons between factories and between device types are like-for-like and not skewed by fleet size.

3. Methodology

The raw JSON was imported directly into Tableau Public. Rather than counting rows, a calculated field was created so the metric maps to something operationally meaningful:

```
Unhealthy = IF [status] = "unhealthy" THEN 10 ELSE 0 END
```

Each unhealthy reading is scored as 10 minutes of downtime (matching the ~10-minute reporting interval), and summing this field gives total estimated downtime in minutes for any slice of the data. Two views were then built on top of this measure:

- **Down Time per Factory** — SUM(Unhealthy) by factory, to compare sites.
- **Down Time per Device Type** — SUM(Unhealthy) by device type, to compare machine categories across all sites.

Both views were combined into a single Tableau dashboard so factory-level and machine-level risk can be read side by side.

4. Dashboard

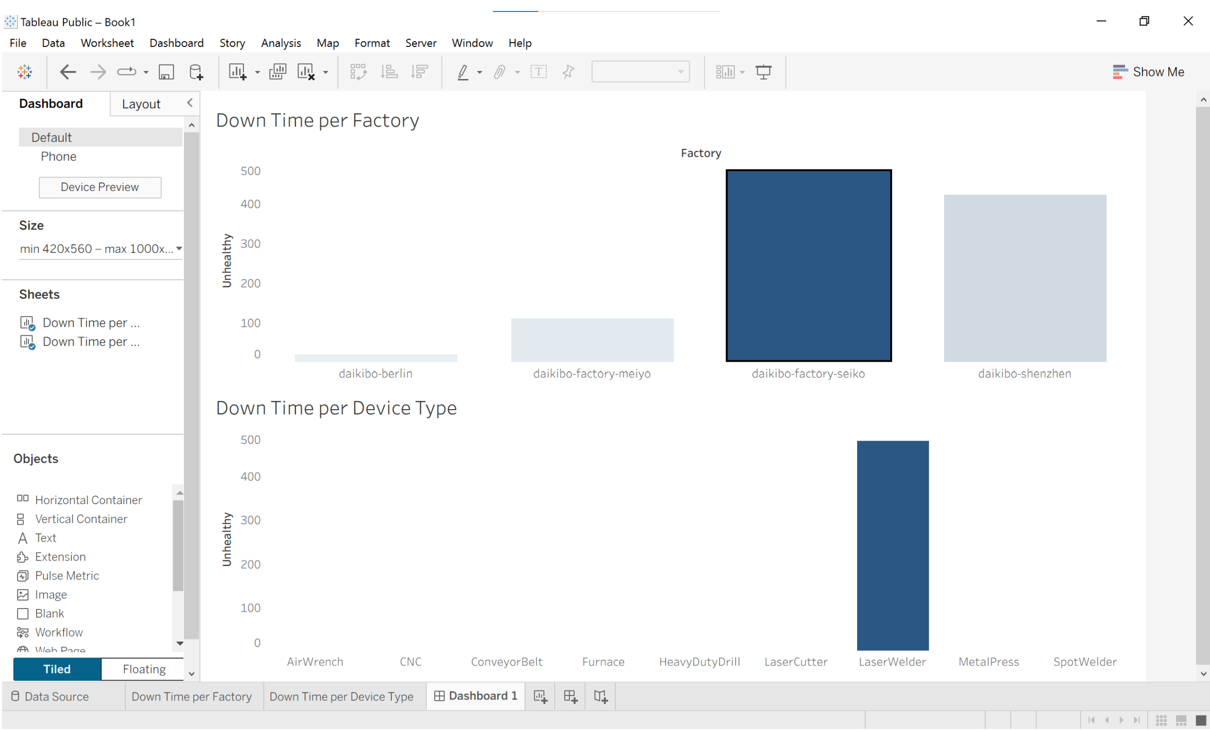

Fig. 1 — Down Time per Factory and Down Time per Device Type, built in Tableau Public.

Note: this figure is a raw application screenshot (captured for this write-up) rather than an exported chart image — see the “Things to double-check” section for a suggested cleanup before publishing.

5. Findings

Across the 9-day sample, devices reported **103 unhealthy readings** out of 160,704 total (0.06%), equivalent to an estimated **1,030 minutes (~17.2 hours)** of downtime fleet-wide. That downtime is heavily concentrated rather than evenly spread:

5.1 By factory

Factory	Country	Unhealthy readings	Est. downtime	Share of total
daikibo-factory-seiko	Japan	48	480 min (8.0 hrs)	46.6%
daikibo-shenzhen	China	42	420 min (7.0 hrs)	40.8%
daikibo-factory-meiyo	Japan	11	110 min (1.8 hrs)	10.7%
daikibo-berlin	Germany	2	20 min (0.3 hrs)	1.9%

daikibo-factory-seiko (Japan) and **daikibo-shenzhen** (China) together account for **87%** of all recorded downtime, despite running the same fleet size as the other two sites. daikibo-berlin is, by comparison, effectively trouble-free in this window.

5.2 By device type

Device type	Unhealthy readings	Est. downtime	Primarily driven by
LaserWelder	48	480 min	daikibo-factory-seiko (all 48)
LaserCutter	43	430 min	daikibo-shenzhen (39), daikibo-factory-meiyo (4)
HeavyDutyDrill	7	70 min	daikibo-factory-meiyo
Furnace	2	20 min	daikibo-berlin
SpotWelder / CNC / ConveyorBelt	1 each	10 min each	daikibo-shenzhen
AirWrench / MetalPress	0	0 min	—

Downtime is not spread across the fleet either: **LaserWelder** and **LaserCutter** alone account for **91%** of all unhealthy readings, and in each case the problem is dominated by a single device at a single factory rather than the device type being unreliable everywhere. In other words, this looks less like “laser machines are generally unreliable” and more like “two specific machines are the problem” — one LaserWelder at daikibo-factory-seiko and one LaserCutter at daikibo-shenzhen together account for the large majority of fleet-wide downtime.

6. Recommendations

- **Prioritise inspection of the two repeat-offender machines** (the LaserWelder at daikibo-factory-seiko and the LaserCutter at daikibo-shenzhen) before scheduling broader fleet maintenance — they represent the fastest route to reducing total downtime.
- **Investigate daikibo-berlin's maintenance practices as a positive benchmark** — with the same fleet size and near-zero downtime, its processes may be worth replicating at the other three sites.
- **Extend the monitoring window** beyond 9 days before drawing firm conclusions — a longer sample would confirm whether the seiko/shenzhen concentration is a persistent pattern or a short-term spike.

- **Layer in the temperature field** (captured but unused here) as a leading indicator — checking whether unhealthy readings are preceded by rising temperature could enable predictive alerts rather than after-the-fact reporting.

Prepared as a portfolio write-up for the Deloitte Australia Data Analytics Job Simulation (Forage). Daikibo Manufacturing and its telemetry data are fictional, provided as part of the simulation exercise.